

1.Program

```
import re

email = input("Enter Email: ")

password = input("Enter Password: ")

mobile = input("Enter Mobile Number: ")

email_pattern = r'^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$'

if re.fullmatch(email_pattern, email):

    print("Valid Email")

else:

    print("Invalid Email")

password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#%&!])[A-Za-z\d@#%&!]{8,}$'

if re.fullmatch(password_pattern, password):

    print("Strong Password")

else:

    print("Weak Password")

mobile_pattern = r'^[6-9]\d{9}$'

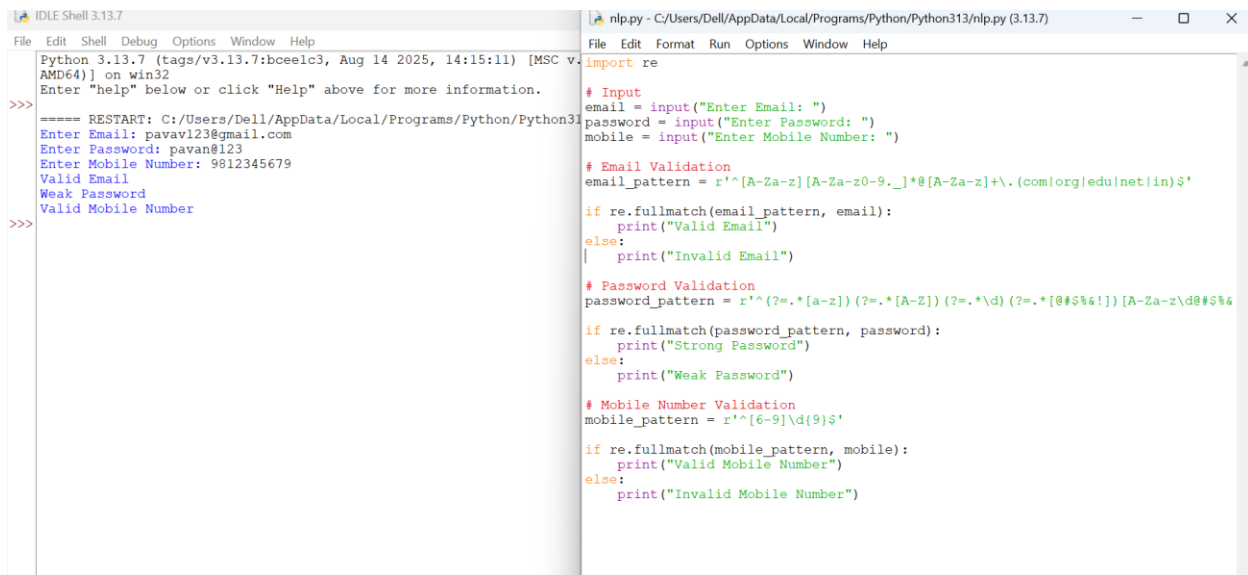
if re.fullmatch(mobile_pattern, mobile):

    print("Valid Mobile Number")

else:

    print("Invalid Mobile Number")
```

Output:



The image shows two side-by-side windows from a Python IDE. The left window, titled 'IDLE Shell 3.13.7', displays the execution output of a program. It shows a restart of the program where the user enters an email 'pavavl23@gmail.com', a password 'pavan@123', and a mobile number '9812345679'. The program outputs 'Valid Email', 'Weak Password', and 'Valid Mobile Number'. The right window, titled 'nlp.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/nlp.py (3.13.7)', shows the source code of the program. The code uses regular expressions to validate email, password, and mobile number inputs.

```
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1916] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313
Enter Email: pavavl23@gmail.com
Enter Password: pavan@123
Enter Mobile Number: 9812345679
Valid Email
Weak Password
Valid Mobile Number
>>>
```

```
nlp.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/nlp.py (3.13.7)
File Edit Format Run Options Window Help
import re

# Input
email = input("Enter Email: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")

# Email Validation
email_pattern = r'^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$'

if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")

# Password Validation
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#%&!]) [A-Za-z\d@#%&]
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")

# Mobile Number Validation
mobile_pattern = r'^[6-9]\d{9}$'

if re.fullmatch(mobile_pattern, mobile):
    print("Valid Mobile Number")
else:
    print("Invalid Mobile Number")
```

2.Program

```
states = input("Enter States (space separated): ").split()

alphabet = input("Enter Input Symbols (space separated): ").split()

transition = {}

print("\nEnter Transition Table:")

for state in states:

    transition[state] = {}

    for symbol in alphabet:

        next_state = input(f"δ({state},{symbol}) = ")

        transition[state][symbol] = next_state

start_state = input("Enter Start State: ")

final_states = input("Enter Final States (space separated): ").split()

n = int(input("Enter Number of Strings: "))

for _ in range(n):

    string = input("\nEnter Input String: ")
```

```

current = start_state

path = [current]

valid = True

for ch in string:

    if ch not in alphabet:

        valid = False

        break

    current = transition[current][ch]

    path.append(current)

print("Transition Path:")

print(" -> ".join(path))

if valid and current in final_states:

    print("Accepted")

else:

    print("Rejected")

```

Output:

The screenshot displays a Python IDE with two panes. The left pane shows the execution of a finite state machine program, and the right pane shows the source code.

Left Pane (Execution Output):

```

Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>
===== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313/nlp
Enter States (space separated): q0 q1 q2
Enter Input Symbols (space separated): a b

Enter Transition Table:
5(q0,a) = q1
5(q0,b) = q0
5(q1,a) = q1
5(q1,b) = q2
5(q2,a) = q1
5(q2,b) = q0
Enter Start State: q0
Enter Final States (space separated): \q2
Enter Number of Strings: 2

Enter Input String: ab
Transition Path:
q0 -> q1 -> q2
Accepted

Enter Input String: aaab
Transition Path:
q0 -> q1 -> q1 -> q1 -> q2
Accepted
>>

```

Right Pane (Source Code):

```

nlp.py - C:/Users/Dell/AppData/Local/Programs/Python/Python313/nlp.py (3.13.7)
File Edit Format Run Options Window Help

states = input("Enter States (space separated): ").split()
alphabet = input("Enter Input Symbols (space separated): ").split()

transition = {}

print("\nEnter Transition Table:")
for state in states:
    transition[state] = {}
    for symbol in alphabet:
        next_state = input(f"5({state},{symbol}) = ")
        transition[state][symbol] = next_state

start_state = input("Enter Start State: ")

final_states = input("Enter Final States (space separated): ").split()

n = int(input("Enter Number of Strings: "))

for _ in range(n):
    string = input("\nEnter Input String: ")

    current = start_state
    path = [current]
    valid = True

    for ch in string:
        if ch not in alphabet:
            valid = False
            break
        current = transition[current][ch]
        path.append(current)

    print("Transition Path:")
    print(" -> ".join(path))

    if valid and current in final_states:
        print("Accepted")
    else:
        print("Rejected")

```

3.Program

```
import re
text = """
Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing
machine learning
"""

while True:

    print("\n----- MENU -----")
    print("1. Search Date")
    print("2. Search Phone Number")
    print("3. Search Hashtag")
    print("4. Search Mention")
    print("5. Search Prefix")
    print("6. Search Suffix")
    print("7. Search Word")
    print("8. Exit")

    choice = int(input("Enter Choice: "))

    if choice == 1:

        result = re.findall(r'\b\d{2}^\d{2}^\d{4}\b', text)

        print("Dates:", result)

    elif choice == 2:

        result = re.findall(r'\b[6-9]\d{9}\b', text)

        print("Phone Numbers:", result)
```

```
elif choice == 3:
    result = re.findall(r'#\w+', text)
    print("Hashtags:", result)
elif choice == 4:
    result = re.findall(r'@\w+', text)
    print("Mentions:", result)
elif choice == 5:
    prefix = input("Enter Prefix: ")
    result = re.findall(r'\b' + re.escape(prefix) + r'\w*', text)
    print("Matching Words:", result)
elif choice == 6:
    suffix = input("Enter Suffix: ")
    result = re.findall(r'\b\w*' + re.escape(suffix) + r'\b', text)
    print("Matching Words:", result)
elif choice == 7:
    word = input("Enter Word: ")
    result = re.findall(r'\b' + re.escape(word) + r'\b', text)
    print("Word Found:", result)
elif choice == 8:
    print("Program Ended")
    break
else:
    print("Invalid Choice")
```

Output:

```
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Dell/AppData/Local/Programs/Python/Python313
----- MENU -----
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit
Enter Choice: 1
Dates: ['12/09/2026']

----- MENU -----
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit
Enter Choice: 2
Phone Numbers: ['9876543210']

----- MENU -----
1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
8. Exit

import re
text = """
Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing
machine learning
"""

while True:

    print("\n----- MENU -----")
    print("1. Search Date")
    print("2. Search Phone Number")
    print("3. Search Hashtag")
    print("4. Search Mention")
    print("5. Search Prefix")
    print("6. Search Suffix")
    print("7. Search Word")
    print("8. Exit")

    choice = int(input("Enter Choice: "))

    if choice == 1:
        result = re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text)
        print("Dates:", result)

    elif choice == 2:
        result = re.findall(r'\b[6-9]\d{9}\b', text)
        print("Phone Numbers:", result)

    elif choice == 3:
        result = re.findall(r'#\w+', text)
        print("Hashtags:", result)

    elif choice == 4:
        result = re.findall(r'@\w+', text)
        print("Mentions:", result)
```